Recitation 9

Two-Sample Tests

Review and Practice Problems

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Goal: Compare an observed difference between two groups with the variation we would expect if their population parameters were equal.

Review: Antetokounmpo's free throws

1. Turn the question into a model. The lecture handout reports the following free throws:

Location	Made	Missed	Attempts
Home (H)	34	10	44
Away (A)	22	19	41

Let $\widetilde{x}_{H,i} \sim \text{Bernoulli}(\theta_H)$ and $\widetilde{x}_{A,j} \sim \text{Bernoulli}(\theta_A)$ indicate made shots. Assume independent shots, with a constant success probability within each location. The parameters θ_H, θ_A are fixed but unknown. For "Is his free-throw percentage *different* at home and away?", use

$$H_0 : \theta_H = \theta_A = \theta, \qquad H_1 : \theta_H \neq \theta_A.$$

This is a **two-sample** problem: both success probabilities are unknown and both are estimated from data. For Bernoulli data, equal means also imply equal distributions; this is not true for arbitrary distributions.

2. Measure the observed difference. Let X_H and X_A denote the observed home and away datasets. Then

$$m(X_H) = \frac{34}{44} \approx 0.773, \qquad m(X_A) = \frac{22}{41} \approx 0.537,$$
$$d_{\text{obs}} = m(X_H) - m(X_A) \approx 0.236.$$

The gap is **23.6 percentage points**. Is it large relative to sampling variation? The counts alone do not answer that question.

3. Work out the variation under the null. Before observing the data, let $\widetilde{m}_H = 44^{-1} \sum_i \widetilde{x}_{H,i}$, $\widetilde{m}_A = 41^{-1} \sum_j \widetilde{x}_{A,j}$, and $\widetilde{d} = \widetilde{m}_H - \widetilde{m}_A$. Under H_0, independence gives

$$\mathbb{E}_0[\widetilde{d}] = 0, \qquad \text{Var}_0(\widetilde{d}) = \theta(1-\theta)\left(\frac{1}{44} + \frac{1}{41}\right).$$

The variances add even though the means are subtracted. The standard error is the square root of this variance.

Review: from a difference to a decision

4. Estimate the null standard error and standardize. Under H_0, all 85 attempts share one success probability. Estimate it by pooling the successes and attempts:

$$\theta_{\mathrm{ML},0} = \frac{34+22}{44+41} = \frac{56}{85} \approx 0.659, \qquad \mathrm{se}_{0,\mathrm{est}} = \sqrt{\theta_{\mathrm{ML},0}(1-\theta_{\mathrm{ML},0})\left(\frac{1}{44}+\frac{1}{41}\right)} \approx 0.1029.$$

Thus $z_{\mathrm{obs}} = d_{\mathrm{obs}}/\,\mathrm{se}_{0,\mathrm{est}} \approx 2.295$: the observed gap is about 2.3 estimated null standard errors above zero. For sufficiently large samples, the corresponding random standardized statistic $\widetilde{z}_{\mathrm{test}}$ is approximately $\mathcal{N}(0,1)$ under H_0.

5. Choose the tail that matches the question. Write $\Phi(u) = \mathbb{P}(\widetilde{z} \le u)$ for $\widetilde{z} \sim \mathcal{N}(0,1)$. Choose the alternative and significance level *before* inspecting outcomes.

Alternative	Approximate p value	Reject at $\alpha = 0.05$ if
$\theta_H > \theta_A$	$1 - \Phi(z_{\mathrm{obs}})$	$z_{\mathrm{obs}} \ge 1.645$
$\theta_H < \theta_A$	$\Phi(z_{\mathrm{obs}})$	$z_{\mathrm{obs}} \le -1.645$
$\theta_H \ne \theta_A$	$2[1 - \Phi(\lvert z_{\mathrm{obs}}\rvert)]$	$\lvert z_{\mathrm{obs}}\rvert \ge 1.960$

For the original question, both directions count as evidence:

$$p_{\mathrm{two}} \approx \mathbb{P}(|\widetilde{z}| \ge 2.295) \approx 0.0218 < 0.05.$$

Reject H_0 at the 5% level. A *prespecified* home-is-better alternative would instead give $p_{\mathrm{right}} \approx 0.0109$.

6. Say what the result means. Under the equal-probability model, a standardized gap at least this extreme in either direction has approximate probability 2.2%. This is evidence of a difference; it is *not* a 2.2% probability that H_0 is true. It does not show that crowd taunts caused the difference: location was not randomly assigned, and opponents, fatigue, and game conditions may differ.

Check the approximation. A common rule of thumb asks for at least 10 expected makes and 10 expected misses in each group, using the pooled estimate. Here these are $(28.99, 15.01)$ at home and $(27.01, 13.99)$ away. This supports a normal approximation, but does not establish independence. Shots within a game may be dependent. The nominal false-positive rate is *approximate* for this plug-in z test and relies on the model.

The same idea for other means. For two independent samples of sizes n_A, n_B, with population means μ_A, μ_B and *known* variances σ_A^2, σ_B^2,

$$\mathrm{se}(\widetilde{m}_B - \widetilde{m}_A) = \sqrt{\frac{\sigma_B^2}{n_B} + \frac{\sigma_A^2}{n_A}}, \qquad z_{\mathrm{obs}} = \frac{m(X_B) - m(X_A)}{\sqrt{\sigma_B^2/n_B + \sigma_A^2/n_A}} \quad (H_0 : \mu_B = \mu_A).$$

This is an exact Gaussian test for independent Gaussian observations with known variances; otherwise a CLT may justify an approximation. With unknown variances and small samples, a suitable two-sample t procedure is needed.

Practice problems

Three core problems; Problem 4 is a backup. A calculator is useful. Use $\Phi(1.645) \approx 0.95$, $\Phi(1.960) \approx 0.975$, and $\Phi(2.295) \approx 0.9891$.

Problem 1: Why do we pool the free throws?

Independent work: 8 minutes. Based on the lecture example. Use the home/away data on page 1 and the independent Bernoulli model.

(a) Write the likelihood under $H_0 : \theta_H = \theta_A = \theta$, up to a factor independent of θ. Derive $\theta_{\mathrm{ML},0}$. Why not use the unweighted average of the two observed success proportions?

(b) Starting from $\mathrm{Var}(\widetilde{x}_{H,i}) = \theta(1-\theta)$ under H_0, derive $\mathrm{Var}_0(\widetilde{d})$. Explain both denominators and the plus sign.

(c) Compute the estimated null SE, standardized statistic, and two-sided p value. State the decisions at $\alpha = 0.05$ and $\alpha = 0.01$.

(d) A student treats 22/41 as the *known* away success probability and includes only the home group's variance contribution in the SE. What source of uncertainty is missing?

Problem 2: Comparing treatment and control means

Independent work: 8 minutes. Archived Exercise 11.19. An experiment compares a control group (A) with an independent treatment group (B). Observations are approximately normally distributed, and the standard deviation of an individual measurement in either group is known to be $\sigma = 5$. Each group has 25 observations.

(a) What is the standard error of $\widetilde{m}_B - \widetilde{m}_A$?

(b) For $H_0 : \mu_B = \mu_A$ versus $H_1 : \mu_B > \mu_A$, give the level-0.05 rejection region, both in standardized units and in the original measurement units.

(c) Suppose the test's p value is 0.07. Would you reject at $\alpha = 0.10$? At $\alpha = 0.05$?

(d) What is the level-0.05 rejection region if instead $H_1 : \mu_B \neq \mu_A$? Explain why the cutoff changes.

Discussion: Why is $5/\sqrt{50}$ not the SE in part (a)?

Problem 3: What does the test tell us?

Independent work: 7 minutes. Adapted from archived Exercise 9.26(b)–(d), with extensions to the free-throw example. Mark each statement true or false and justify your answer in one or two sentences. For a false statement, give a correct replacement.

(a) If the p value is 0.03, the test rejects at significance level 0.02.

(b) If a test using the rule "reject when $p \leq \alpha$" rejects at significance level 0.06, then its p value is at most 0.06.

(c) The free-throw test's two-sided p value 0.0218 is the probability that the home and away success probabilities are equal.

(d) If we compute away minus home instead of home minus away, the two-sided p value is unchanged.

(e) After seeing that the home percentage is higher, we may replace the planned two-sided test by a one-sided test and halve its p value while keeping the advertised significance level.

(f) Rejecting equality establishes that crowd taunts caused the lower away success rate.

Problem 4 (backup): Is the normal approximation credible?

Independent work: 5 minutes. Adapted from archived Recitation Worksheet 11, Problem 2. A study records exposure to an environmental factor and disease status:

	Not exposed	Exposed
No disease	128	7
Disease	19	7

Model individuals as independent, with disease probabilities θ_N and θ_E in the two groups.

(a) State hypotheses for whether the disease probabilities differ, and find the observed exposed-minus-unexposed difference.

(b) Find the pooled null estimate and the expected disease/no-disease counts in each group using that estimate.

(c) Does the "at least 10 expected successes and failures per group" guideline hold? Explain why computing a normal-tail p value alone would not settle whether the approximation is reliable.